



ELEMENTARY MATH (CALCULUS)



INTEGRATION TECHNIQUES & APPROACH



LEARNING OBJECTIVES

Upon completion of this unit, you should be able to:

- identify some rules of integration,
- evaluate integrals using the standard formula,
- evaluate integrals using some useful techniques such as substitution, integration by parts, etc.



INTRODUCTION

The problem of calculating the derivative of an elementary function can always be solved by a systematic application of the standard rules. However, the inverse problem of integration is very different and has no such clear-cut solution. In this note, we discuss some standard techniques for computing integrals: substitution methods, integration by parts, trigonometric substitution, and partial fractions.



STANDARD INTEGRALS

Here is a list of common indefinite integrals that should already be familiar:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$$

$$\int x^{-1} dx = \ln(x) + C$$

$$\int e^x dx = e^x + C$$

$$\int \sin(x) dx = -\cos(x) + C$$

$$\int \cos(x) dx = \sin(x) + C$$



STANDARD INTEGRALS

$$\int \sec^2(x) dx = \tan(x) + C$$

$$\int \sec(x) \tan(x) dx = \sec(x) + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} + C$$

$$\int \frac{1}{1+x^2} dx = \tan^{-1} + C$$

$$\int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1} + C$$



STANDARD INTEGRALS

$$\int \ln(x)dx = x\ln(x) - x + C$$

$$\int \tan(x)dx = -\ln(\cos(x)) + C$$

$$\int \sec(x)dx = \ln(\sec(x) + \tan(x)) + C$$

$$\int \csc(x)dx = -\ln(\csc(x) + \cot(x)) + C$$

$$\int \cot(x)dx = \ln(\sin(x)) + C$$

In what follows, we present some useful integration techniques.



SUBSTITUTION

The general substitution formula states that $\int f'(g(x)) \cdot g'(x) dx = f(g(x)) + C$. It is just the Chain Rule, written in terms of integration via the Fundamental Theorem of Calculus. This substitution is usually done in the following sense.

Example 1.

Evaluate $\int_0^2 (x+2)^3 dx$.

Solution

Let $u = (x+2)$, so that $du = dx$. Notice that if $x = 0$, then $u = 2$, and if $x = 2$, then $u = 4$. Therefore,

$$\int_0^2 (x+2)^3 dx = \int_2^4 u^3 du = \frac{1}{4} u^4 \Big|_2^4 = \frac{1}{4} (4^4 - 2^4) = 60$$



SUBSTITUTION

Example 2.

Evaluate $\int 2xe^{x^2} dx$.

Solution

Since the exponential component is x^2 , we let $u = x^2$, so that $du = 2xdx$. Therefore,

$$\int 2xe^{x^2} dx = \int e^{x^2} 2xdx = \int e^u du = e^u + C$$



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TASK

Evaluate $\int_0^1 xe^x dx$



INTEGRATION BY PARTS

Let $f(x) = u$ and $g(x) = v$ then we have
 $\int u dv = uv - \int v du$.

Example 1.

Evaluate $\int x e^x dx$.

Solution:

Let $u = x$. Then $du = dx$. Also, let $dv = e^x dx$.
Then $v = e^x$. Therefore, we have

$$\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C$$



INTEGRATION BY PARTS

Example 2.

Evaluate $\int 3x^2 \sin(x) dx$.

Solution:

Let $u = 3x^2$. Then $du = 6x dx$. Also, let $dv = \sin(x) dx$. Then $v = -\cos(x)$. Therefore, we have

$$\begin{aligned}\int 3x^2 \sin(x) dx &= 3x^2(-\cos(x)) - \int -\cos(x) (6x dx) \\ &= -3x^2 \cos(x) + 6 \int x \cos(x) dx\end{aligned}$$

To again evaluate $\int x \cos(x) dx$, let $u = x$ and $dv = \cos(x) dx$. Then $du = dx$, $v = \sin(x)$, so that

$$\int x \cos(x) dx = x \sin(x) - \int \sin(x) dx = x \sin(x) + \cos(x)$$



INTEGRATION BY PARTS

Finally, we have

$$\int 3x^2 \sin(x) dx = -3x^2 \cos(x) + 6(x \sin(x) + \cos(x)) + C$$



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TASK

Evaluate $\int x \ln(x) dx$



TRIGONOMETRIC SUBSTITUTION

To evaluate the following type of integrals, some trigonometric substitutions are required.

Example 1.

Evaluate $\int \sqrt{1-x^2} dx$

Solution

Here, we let $x = \sin(u)$, so that $dx = \cos(u) du$. Hence, our integral becomes

$$\begin{aligned}\int \sqrt{1-x^2} dx &= \int \sqrt{1-\sin^2(u)} \cos(u) du \\ &= \int \sqrt{\cos^2(u)} \cos(u) du = \int \sqrt{\cos^2(u)} \cos(u) du = \int \cos^2(u) du\end{aligned}$$



TRIGONOMETRIC SUBSTITUTION

Noting that $\cos^2(u) = \frac{1+\cos(2u)}{2}$, we have

$$\int \cos^2(u) du = \frac{u}{2} + \frac{\sin(2u)}{4} + C$$

Setting back $u = \sin^{-1}(x)$, we obtain

$$\int \sqrt{1-x^2} dx = \frac{\sin^{-1}(x)}{2} + \frac{\sin(2\sin^{-1}(x))}{4} + C$$

To further simplify this, notice that $\sin(2u) = 2\sin(u)\cos(u) = 2\sin(\sin^{-1}(x))\sqrt{1-\sin^2(u)} = 2x\sqrt{1-\sin^2(\sin^{-1}(x))} = 2x\sqrt{1-x^2}$. Hence,

$$\int \sqrt{1-x^2} dx = \frac{\sin^{-1}(x)}{2} + \frac{x\sqrt{1-x^2}}{2} + C.$$



TRIGONOMETRIC SUBSTITUTION

Example 1.

Evaluate $\int \sqrt{4 - 9x^2} dx$

Solution

Let us rewrite the integral in the following manner

$$\int \sqrt{4 - 9x^2} dx = \int \sqrt{4(1 - (3x/2)^2)} dx = \int 2\sqrt{1 - (3x/2)^2} dx.$$

Now, let $3x/2 = \sin(u)$, so that $dx = (2/3)\cos(u) du$. Hence, our integral becomes

$$\begin{aligned} \int 2\sqrt{1 - (3x/2)^2} dx &= \int 2\sqrt{1 - \sin^2(u)} (2/3)\cos(u) du \\ &= \frac{4}{3} \int \cos^2(u) du = \frac{4u}{6} + \frac{4\sin(2u)}{12} + C \end{aligned}$$



TRIGONOMETRIC SUBSTITUTION

By substituting the value of u and further simplifying, we have

$$\begin{aligned} & \int \sqrt{4 - 9x^2} dx \\ &= \frac{2\sin^{-1}(3x/2)}{3} + \frac{2\sin(\sin^{-1}(3x/2))\cos(\sin^{-1}(3x/2))}{3} + C \\ &= \frac{2\sin^{-1}(3x/2)}{3} + \frac{x\sqrt{4 - 9x^2}}{2} + C \end{aligned}$$



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TASK

Evaluate $\int \sqrt{1+x^2} dx$



PARTIAL FRACTIONS

Some integrals are evaluated by first resolving them into partial fractions.

Example 1.

Evaluate $\int \frac{1}{x^2 + 3x} dx$.

Solution

Factoring $\frac{1}{x^2 + 3x}$, we get $\frac{1}{x(x+3)}$, so that $\frac{1}{x(x+3)} = \frac{A}{x} + \frac{B}{(x+3)}$. By clearing the denominator, we have $1 = A(x+3) + B(x)$. Setting $x = 0$ and $x = -3$ yield $A = \frac{1}{3}$, $B = -\frac{1}{3}$. Therefore,

$$\int \frac{1}{x^2 + 3x} dx = \int \left(\frac{1/3}{x} - \frac{1/3}{x+3} \right) dx = \frac{1}{3} \ln(x) - \frac{1}{3} \ln(x+3) + C$$



PARTIAL FRACTIONS

Example 2.

Evaluate $\int \frac{7x-6}{(x-2)(x+3)} dx$.

Solution

Resolving into partial fractions, we have $\frac{7x-6}{(x-2)(x+3)} = \frac{A}{(x-2)} + \frac{B}{(x+3)}$. By clearing the denominator, we have $7x - 6 = A(x + 3) + B(x - 2)$, so that $A = \frac{8}{5}, B = \frac{27}{5}$.

Therefore,

$$\int \frac{7x-6}{(x-2)(x+3)} dx = \int \left(\frac{8/5}{x-2} + \frac{27/5}{x+3} \right) dx = \frac{8}{5} \ln(x-2) + \frac{27}{5} \ln(x+3) + C$$



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TASK

Evaluate $\int \frac{5x+7}{(x-1)(x+3)} dx$.



SUMMARY

Some standard techniques for computing integrals of a function have been discussed.

There are standard integrals for some well-known functions. The general rule for integrating a basic function is given by

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1.$$

There are various techniques and approach for evaluating integrals, including substitution methods, integration by parts, trigonometric substitution, and partial fractions.



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FURTHER READING

" Calculus Know-It-ALL " by Stan Gibilisco.

" Theory and Problems of ADVANCED CALCULUS" by
Robert Wrede, and Murray R. Spiegel.

“Advanced Engineering Mathematics" by H. K. Dass.

“Engineering Mathematics" by K. A. Stroud.



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THANK
YOU